

Intro to Software Engineering – Quiz #3

June 25, 2024

Quiz #2
Total: 50 points
Due Date: July 10th, 2024

*This quiz must be completed individually.
I.e. you should work alone without any help or collaboration from others.
Please provide detailed answers to the following exercises.*

Question 1. [5 marks]

You have a MongoDB collection named **myCollection** that stores documents with various attributes including **name**, **age**, and **profession**. Write a MongoDB query to retrieve documents from this collection where the **profession** is either "Software Engineer" or "Data Scientist" and the individual is older than 25 years. Use the **AND** and **OR** logical operators to combine the conditions.

Question 2. [5 marks]

You are tasked with retrieving information from a MongoDB collection named `peaks`, which documents details about mountain peaks. Each document includes fields such as `name`, `height`, `location`, and a nested document `details` that contains `ascents` and `prominence`.

Construct a MongoDB query to find mountain peaks that satisfy all the following criteria:

1. The peaks are located in both "China" and "Nepal", and the location order does not matter.
2. The peaks have a height greater than 8000 meters.
3. The peaks have a prominence greater than 1000 meters.

Question 3 [10 marks]

Given a MongoDB collection `oceanicRecords`, each document includes a nested document `observations` among other fields like `species`, `depth`, and `location`. Construct a MongoDB query to extract only the following information for each sea creature observed during various oceanic expeditions:

1. The name of the species (`species`).
2. The depth at which the species was observed (`depth`).
3. The date when the species was last seen (`lastSeen` from the nested `observations` document).
4. The total number of sightings (`sightingsCount` from the nested `observations` document).

Additional requirements:

- Exclude the `_id` field from the results..
- Write your query as if you are aiming to optimize performance by limiting the data returned from the database.

Question 4 [10 marks]

Given a MongoDB collection `cityEvents`, which stores details about various events including `eventName`, `city`, `date`, and `attendees`, construct a MongoDB query based on the following requirements:

1. Identify events that are either in "New York" or "Los Angeles" and have more than 100 attendees.
2. Sort the results by `date` in descending order to get the most recent events first.
3. Limit the output to only the top 5 results.
4. Return only the `eventName` and `city` for each document, excluding all other fields including the default `_id` field.
5. Make sure to use MongoDB's aggregation framework. (*`db.cityEvents.aggregate`*)

Question 5 [5 marks]

1. **products** Collection:

- **_id**: A unique identifier for each product.
- **name**: The name of the product.
- **price**: The price of each product.
- **category**: The category to which the product belongs (e.g., electronics, groceries).
- **quantity**: The available quantity of each product.

Count the Number of Products in Each Category

Question 6 [10 marks]

Nowadays, many features of NoSQL databases are implemented inside SQL databases, like the implementation of JSONB in PostgreSQL to support unstructured data.

Here is an example:

```
CREATE TABLE sample_jsonb (  
  id serial NOT NULL PRIMARY KEY,  
  data jsonb  
);
```

```
INSERT INTO sample_jsonb (data) VALUES  
('{"name": "First", "count": 12, "orderDate": "2022-03-03T12:14:01", "extra": "some text"}');  
INSERT INTO sample_jsonb (data) VALUES  
('{"name": "Second", "count": 23, "orderDate": "2022-04-03T15:17:01"}');
```

```
SELECT data->>'name', data->>'orderDate'  
FROM sample_jsonb  
WHERE (data->'count')::int > 12;
```

We can also define indexes for this.

Now answer these questions to the best of your knowledge.

- a) Why do people still use MongoDB or similar databases? Why should someone choose NoSQL over SQL?
- b) On the other hand, why do some companies nowadays are changing their tech stack from NoSQL to SQL?

Question 7 [5 marks]

Explain what this query is doing?

```
db.products.aggregate([
  { $group: {
    _id: "$category",
    totalQuantity: { $sum: "$quantity" },
    averagePrice: { $avg: "$price" }
  }},
  { $match: { averagePrice: { $gt: 50 } } }
]);
```